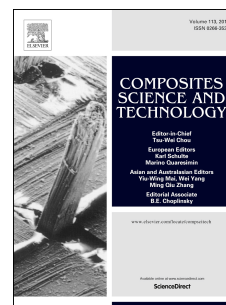


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Improved Dielectric Properties of Polypropylene-based Nanocomposites via Co-filling with Zinc Oxide and Barium Titanate

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Abstract: Tetra-needle-shaped zinc oxide whiskers (T-ZnOw) and barium titanate (BT) nanoparticles were embedded into polypropylene (PP) matrix to construct ternary nanocomposites with improved dielectric properties. The nanocomposites were fabricated via a melt blending and subsequent compression molding approach. The microstructure, crystallization behavior, and dielectric properties of as-fabricated nanocomposites were investigated in detail. Results reveal that the selecting processing approach could afford isotropic nanocomposites with T-ZnOw and BT nanoparticles homogeneously dispersed throughout the PP matrix. The T-ZnOw/PP binary composites exhibited typical characteristics of percolation system with a relatively small threshold of 13 vol%. When the content of T-ZnOw approaching the percolation threshold, the dielectric constant and loss tangent of the binary composites were dramatically enhanced. After introducing BT nanoparticles, the resultant ternary composites showed further improved dielectric performance. For the ternary T-ZnOw/BT/PP with the BT content of 15 vol% and T-ZnOw content of 9.76 vol%, dielectric constant and loss tangent at 1 kHz reached 11 and 0.04, respectively.

Keywords: Polypropylene; Composites; Dielectric Properties.

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1. Introduction

Dielectrics which can store/dissipate energy in an applied electric field, are one type of fundamental materials for modern electronic devices and electrical systems such as capacitors, transistors, and cables, etc.[1-7] Over the past decade, dielectrics with high volumetric energy density have drawn tremendous interests as the rapid development of integrated circuits for smart electronics. The energy density of dielectrics can be roughly estimated by $\varepsilon_0 \varepsilon_r E_b^2 / 2$, where ε_0 is vacuum dielectric constant, ε_r is the relative dielectric constant, and E_b is the breakdown strength. Thus to obtain higher energy density, developing dielectrics with high ε_r and high E_b is the key. Currently, biaxially-oriented polypropylene (BOPP) films have been widely applied in organic thin film capacitors due to their advantages of high E_b (~ 600 MV/m), high quality factor ($\sim 10^3$), excellent mechanical flexibility, easy processing, and low cost.[8] However, the relatively low dielectric constant of PP (~ 2.2) severely limits the realization of high energy density. Therefore, it is very challenging to fabricate PP dielectric films with improved ε_r while avoiding deterioration of other properties including E_b , dielectric loss, mechanical flexibility, etc.

Recently, developing dielectrics with high energy density through the compositing strategy has attracted tremendous interests. According to the nature of fillers, the polymer-based composite dielectrics can be roughly classified into two categories: polymers filled with high ε_r ceramics, and polymers filled with conductive particles. For the ceramic/polymer composites, the enhancement in dielectric constant relies on the increase of volume fraction of ceramic fillers following the mixing rules. Practically, ceramic nanoparticles with an ultrahigh ε_r such as barium titanate (BT) are commonly selected as fillers to mix with the polymer matrix.[9-11] However, apparently enhanced ε_r of resultant composites can only be realized under high filler loading (~ 50 vol% or greater), which would result in deteriorated mechanical properties and processability of the composite films. In contrast, composites filled by conductive particles exhibit enhanced ε_r following a percolation law. Ultrahigh ε_r can be expected when the concentration of the conductive filler is approaching the percolation threshold. Various conductive fillers, such as silver, nickel, polyaniline, carbon

nanotubes, and graphene, have been examined to prepare percolative composite systems with high ϵ_r [1,2,12,13]. Although this type of composites processes high ϵ_r , the accompanied high dielectric loss and reduced electrical breakdown strength make it unacceptable for dielectric capacitor applications. To address these challenges, many researchers have focused on using semiconductive fillers such as ZnO, which can be treated as compromised materials of high ϵ_r ceramics and conductive particles.[14-18] Specifically, radial shaped ZnO nanoparticles, due to their particular topological structure and physical properties, have intensively investigated as semiconductive fillers to improve the dielectric performance over past few years. In 2008, Wang, et al. confirmed blending radial ZnO with a high aspect ratio into poly(vinylidene fluoride) (PVDF) matrix can obtain composites with significantly higher ϵ_r and better thermal stability than by using conventional spherical ZnO nanoparticles as well as showing a low percolation threshold.[19] Zha et al. prepared polytetrafluoroethylene (PTFE) based composites by using tetra-needle-shaped ZnO whisker (T-ZnOw) as the filler and examined the dielectric properties and thermal conductivity of resultant composites.[20] They also confirmed the utilization of radial ZnO is beneficial to constructing percolative composites with enhanced ϵ_r and relatively low percolation threshold. Some researchers also reported incorporating high ϵ_r ceramics into the ZnO-filled binary system to further improve the dielectric properties. For example, Wang fabricated radial ZnO/BT/PVDF ternary composites and found that the introduction of BT into ZnO/PVDF binary system can enhance the dielectric constant and decrease dielectric loss further.[21]

The investigation on dielectric properties of radial ZnO-filled PP composites has not been reported in the literature so far, though some reports have dealt with the mechanical performance and thermal stability of ZnO/PP composites.[22] Thus in this work, we present the preparation and dielectric properties of ternary PP-based composites co-filled by T-ZnOw and BT nanoparticles. We found that the introduction of T-ZnOw indeed improved the ϵ_r of PP considerably. While the dielectric loss of T-ZnOw/PP composites also deteriorates dramatically as the increase of T-ZnOw content. To suppress the formation of the T-ZnOw percolative conducting network, spherical BT

nanoparticles were selected as the second filler to blend with T-ZnOw/PP composites. The results demonstrate the deteriorated dielectric loss can be partially restored by incorporating the BT nanoparticles. The microstructure, crystalline behavior, and dielectric performance of the binary T-ZnOw/PP and ternary T-ZnOw/BT/PP composites over different filler loadings were characterized and discussed in detail in the following sections.

2. Experimental

2.1. Materials

Isotactic PP pellets (T30S) with a melt flow index of 3 g/10 min were purchased from Maoming Petroleum Chemical Co. Ltd. (China). T-ZnOw was obtained from Chengdu Crystrealm Co. Ltd. (China). BT nanoparticles with an average diameter of 100 nm were purchased from Shandong Sinoceramic Functional Materials (China). All other reagents and solvents were commercial analytical grade products and used without further purification.

2.2. Fabrication of PP-based nanocomposite films

The PP-based binary and ternary nanocomposites films were fabricated through a melt compounding and subsequent compression molding approach. Firstly, desired amount of T-ZnOw and BT were measured and premixed with PP pellets. The masterbatch was then fed into a Haake miniLab micro-compounder (Thermo Fisher Scientific, USA) and melt compounded at 175 °C with a screw speed of 60 rpm for 10 min. Finally, the nanocomposites were compression molded into a circular die at 190 °C under a pressure of 10 MPa for 30 min and cooled gradually to room temperature.

2.4. Characterization

The morphology of the fillers and fracture surfaces of nanocomposites were observed by using a field emission scanning electron microscope (SEM, HITACHI S4700, Japan) with an accelerating voltage of 10 kV. The samples were sputter-coated with gold before observation. X-ray diffraction (XRD) patterns of fabricated nanocomposites were collected by using a D/MAX 2500 system (Rigaku, Japan) with a Cu K α radiation source (40 KV, 150 mA). The crystalline behavior of

T-ZnOw/PP and T-ZnOw/BT/PP nanocomposites were characterized by using a differential scanning calorimeter (DSC-60, SHIMADZU, Japan). The samples were heated from 50°C to 220°C with a heating rate of 10°C/min under nitrogen atmosphere and maintained at this temperature for 1 min; then the samples were cooled down to 50°C at the cooling rate of 10°C /min. Dielectric properties of the T-ZnOw/PP and T-ZnOw/BT/PP composites were measured by using an Agilent 4294A impedance analyzer. Thin film samples with the area of about 1 cm², onto which silver electrodes had been painted, were tested.

3. Results and Discussion

The structure of the nano-fillers, T-ZnOw and BT nanoparticles, was firstly characterized by using SEM observation and XRD. The radial tetra-needle shape of ZnO whisker can be clearly seen in Figure 1a. The length of each needle is in the range of 5-10 μm , whereas the diameter of each needle at the joint point is about 1-3 μm . The morphology of BT nanoparticles is in the spherical shape with the diameter of about 100-150 nm. Some aggregates composed of several BT nanoparticles can also be observed in Figure 1b. XRD patterns of T-ZnOw and BT nanoparticles are shown in Figure 1c. The diffraction peaks are labeled with the corresponding crystalline planes. The results reveal that the T-ZnOw holds a hexagonal wurtzite structure [23] while the BT nanoparticles are cubic phase perovskites [24].

The dispersion of T-ZnOw with various contents in PP matrix was also investigated by SEM. As revealed in Figure 2, all the T-ZnOw/PP binary composites show the relatively homogeneous dispersion of T-ZnOw. While the tetra-needle shape of T-ZnOw can hardly be observed on the fractured surfaces, suggesting that most of the T-ZnOw were broke off into single needles during the shear mixing process. The fractured surfaces of the composites are rather smooth, due to the brittle nature of PP matrix. Moreover, the interfaces between the ZnO needles and PP matrix are compact, demonstrating the interfacial interactions are strong enough. However, some voids and cracks can be seen in the interfacial region when the volume fraction of T-ZnOw reached a high value (as illustrated in Panels a5 and a6 of Figure 2).

The morphology of T-ZnOw/BT/PP ternary composites is shown as Panel b1-b3 in Figure 2. Those three samples contain the same volume fraction of T-ZnOw (9.76 vol%) while different contents of BT. It can be seen that the BT particles in nanometer scale and the ZnO needles in micrometer scale are uniformly dispersed throughout the PP matrix. Compared to the fractured surfaces of T-ZnOw/PP binary composite, the T-ZnOw/BT/PP ternary composites exhibit smoother sectional morphology with fewer cracks.

The effects of T-ZnOw on the non-isothermal crystallization behavior of PP were analyzed by using DSC. Figure 3 gives typical melting and crystallization curves of T-ZnOw/PP binary composites and T-ZnOw/BT/PP ternary composites under a heating/cooling rate of 10 °C/min. From Figure 3a it can be observed that the melting onset temperature gradually decreases as the increasing of T-ZnOw content. The cooling curves of T-ZnOw/PP are quite different from neat PP as shown in Figure 3b. The crystallization onset peaks of the binary composites shift to higher temperature region by 5 °C. All these observations confirm that the ZnO needles embedded throughout the PP matrix are acting as nucleation agents which can facilitate the formation of PP crystals with smaller domain size. Compared to the cooling curve of T-ZnOw/PP, the crystallization behavior changed a lot after the introduction of the third phase of BT nanoparticles. As shown in Figure 3d, new crystallization peaks appeared at the high-temperature side for the ternary composites, which may be caused by a change in crystal form of PP or a change in the size of the crystal domain, or both. To examine if there is any structural transformation in crystalline forms of PP, the XRD patterns of pure PP, T-ZnOw/PP binary composite, and T-ZnOw/BT/PP ternary composite were measured. As depicted in Figure 4, only monoclinic α -form of isotactic PP with typical diffractions at $2\theta = 14.1, 16.9, 18.6, 21.2$ and 21.9 degrees can be observed for all the three samples. Meanwhile, the characteristic diffraction peak corresponding to hexagonal β -form of PP, which should appear at $2\theta = 16.1$ degree, is always negligible. These results reveal the filling of T-ZnOw and BT nanoparticles under current processing conditions cannot induce the crystalline transformation of PP. The crystalline structure of PP in the as-fabricated composites is primarily in

the conventional monoclinic α -form. Therefore, the new peaks emerged on DSC cooling curves of T-ZnOw/BT/PP ternary composites should be ascribed to small size α -crystals of PP nucleated by BT nanoparticles. The crystallinity (X_c) of PP in the composites was calculated from the crystallization curves by using,

$$X_c = \frac{\Delta H_c}{\Delta H_m^0} \quad (1)$$

where ΔH_c is the crystallization enthalpy of PP extracted from the DSC curves and ΔH_m^0 is the melting enthalpy of a 100% crystalline PP sample. The value was taken as 177.0 J/g according to literature [25]. The calculation results were summarized in Table 1. It can be clearly observed that the crystallinity of PP improved apparently after introducing T-ZnOw and BT nanoparticles.

Figure 5a and 5b show the dependence of dielectric constant and loss tangent of T-ZnOw/PP binary composites on the frequency with different T-ZnOw filling contents. As illustrated in the zoom-in figures, the frequency dependence of dielectric constant and loss tangent over the measured frequency range (1 kHz-1 MHz) is not apparent when the volume fraction of the semiconductive filler is less than 6.50 vol%. The values of dielectric constant and loss tangent increase slowly along with the increasing filler content. The dielectric behavior of such composites is similar to the conventional polymer composites filled with high ϵ_r ceramics. However, when the volume fraction of T-ZnOw goes beyond 6.50 vol%, the dielectric behavior of the composites begins frequency dependent. Both the ϵ_r and the loss tangent values increase remarkably as the increasing of T-ZnOw loading. The dielectric constant and loss tangent of T-ZnOw/PP binary composites at 1 kHz are summarized in Figure 6a. Both the values of dielectric constant and loss tangent are improved by nearly one order of magnitude when the fraction of T-ZnOw increases from 10 to 15 vol%. The dielectric behavior is in good agreement with the percolation system. For the polymer composites filled with conductive additives, the enhancement of the dielectric constant can be predicted by using the percolation power law, which can be expressed as follows [13]:

$$\epsilon_{\text{eff}} \propto \epsilon_m (f_c - f)^s \text{ for } f < f_c \quad (2)$$

where ε_{eff} and ε_m are the dielectric constant of the resultant composite and the polymer matrix, f is the volume fraction of the filler, f_c is the percolation threshold and s is the critical exponent. Figure 6b exhibits the best fits of the dielectric constant of T-ZnOw/PP binary composites to the equation (2). Results reveal that the percolation threshold is $f_c = 13$ vol% with the critical exponent of $s = -0.88$.

Though the dielectric constant of PP can be enhanced by introducing the T-ZnOw, the dielectric loss also increases largely. As discussed in the Introduction part, the high dielectric loss is undesired for the thin capacitor applications. Therefore, the BT nanoparticles were further incorporated into the binary composites to modulate the dielectric properties. To investigate how the third BT component affects the dielectric properties of composites, various contents of BT were mixed with T-ZnOw/PP composites with the constant volume fraction of T-ZnOw (9.76 vol%). As shown in Figure 7a, the dielectric constant of T-ZnOw/PP composites could be further improved by adding BT nanoparticles. The particle size of BT is largely different from the T-ZnOw. The nanometer-sized BT particles can be filled in the gaps of large ZnO needles, which is conducive to increase of total filling quantity and increase the interface area in composite material at the same time, thereby increasing dielectric constant. Dang et al. have studied the effect of the diameter of BT particles on the dielectric constant of composite dielectrics.[26] By selecting suitable volume ratio of micro/nano BT, the dielectric properties of micro/nano co-filled composites are far better than those filled by using only one single type of micro/nanometer fillers. For example, the PVDF-based composite film co-filled with micro/nanosized BT showed ε_r as high as 52 at 100 Hz, which is about two times higher than that of PVDF filled with the same volume fraction of BT nanoparticles.[26] Similarly, the results present here also confirm that improved dielectric constant and reduced dielectric loss of composites can be achieved by co-filling with micrometer-scale ZnO

needles and nanometer-scale BT nanoparticles. An important finding is the loss tangent of T-ZnOw/PP decreases as the introduction of BT nanoparticles (as shown in Figure 7b). This should be attributed to BT nanoparticles which act as barriers blocking the charge carrier transport of ZnO conductive network. The ϵ_r and the loss tangent of the ternary composites reach 11 and 0.04, respectively, by filling with 15 vol% of BT nanoparticles. As shown in Table 2, these values are comparable to those of PP-based composite dielectrics which are filled with BT nanoparticles or co-filled with BT and other type of nanomaterials. In addition, introducing BT nanoparticles can also reduce the frequency dependence of dielectric properties of T-ZnOw/PP composites, which is welcome for the practical applications.

4. Conclusion

Polypropylene-based composites co-filled with T-ZnOw and BT nanoparticles were successfully prepared by melt blending and subsequent compression molding. The processing approach resulted in the homogenous dispersion of micro-scale T-ZnOw semiconductive fillers and nano-scale BT ceramic particles throughout the PP matrix. The dielectric analysis shows that the addition of T-ZnOw significantly improves the dielectric constant of PP following the percolation law. When approaching the percolation threshold of 13.0 vol%, the loss tangent of resultant T-ZnOw/PP composites deteriorated rapidly. After introducing the third component of BT nanoparticles, the dielectric properties of T-ZnOw/PP composites were further improved. For the ternary composite T-ZnOw/BT/PP with the BT content of 15 vol% and T-ZnOw content of 9.76 vol%, the dielectric constant and loss tangent at 1 kHz reached 11 and 0.04. This type of ternary composite dielectrics may find applications in future flexible thin film capacitors.

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Captions of Tables and Figures

Table 1. DSC data for PP-based composites with various contents of fillers under cooling rate of 10 °C/min.

Table 2. Comparison of dielectric properties of typical composite dielectrics filled with various fillers.

Figure 1. Characterization of T-ZnOw and BT nanoparticles. (a) and (b) are typical SEM images of T-ZnOw and BT nanoparticles, respectively. (c) shows the XRD patterns of T-ZnOw and BT nanoparticles.

Figure 2. SEM images of fractured cross-sections of the nanocomposite films with different filler contents: a1-a6 represent the T-ZnOw/PP binary composites with the T-ZnOw loading of 1.79vol%, 3.90vol%, 6.50vol%, 9.76vol%, 14.3vol%, and 19.5vol%, respectively; b1-b3 represent the ternary nanocomposites with the same T-ZnOw loading of 9.76vol% and different BT loading of 5vol%, 10vol%, and 15vol%.

Figure 3. DSC curves of melting and crystallization of T-ZnOw/PP binary composites and T-ZnOw/BT/PP ternary composites with various amounts of filler loadings. (a) and (c) represent the melting process, while (b) and (d) represent the crystallization process. All the curves are normalized by using the mass of PP in the samples. The filler contents the samples for each curve are denoted in the figures.

Figure 4. Typical XRD patterns of PP, T-ZnOw/PP binary composite with 9.76 vol% of T-ZnOw, and T-ZnOw/BT/PP ternary composite with 9.76 vol% of T-ZnOw and 15 vol% of BT. The diffraction peaks corresponding to the monoclinic α -form of PP are denoted.

Figure 5. Dielectric constant (a) and loss tangent (b) of T-ZnOw/PP composites with different filler contents as a function of the electric field frequency measured at room temperature. (a') and

(b') are the zoom-in figures for (a) and (b), respectively.

Figure 6. (a) Dielectric constant and loss tangent of T-ZnOw/PP composites at 1 kHz as a function of filler volume fraction. (b) Linear fit of experimental dielectric constant data in (a) to Equation (1).

Figure 7. (a) Dielectric constant and (b) loss tangent of T-ZnOw/BT/PP composites with the constant T-ZnOw content of 9.76 vol% as a function of the electric field frequency measured at room temperature.

Table 1.

Sample	vol% of T-ZnOw	vol% of BT	T_c (°C)	ΔH_c (J/g)	X_c (%)
PP	n/a	n/a	115.8	76.5	43.2
T-ZnOw/PP	1.79	n/a	120.9	82.9	46.8
T-ZnOw/PP	3.90	n/a	120.8	86.9	49.1
T-ZnOw/PP	6.50	n/a	119.8	78.9	44.6
T-ZnOw/PP	9.76	n/a	120.5	88.2	49.8
T-ZnOw/PP	14.35	n/a	120.9	89.8	50.7
T-ZnOw/PP	19.50	n/a	120.7	97	54.8
T-ZnOw/BT/PP	9.76	5	133.1, 126.1, 119.9	74.4	42.0
T-ZnOw/BT/PP	9.76	10	132.8, 126.8, 120.1	104.9	59.2
T-ZnOw/BT/PP	9.76	15	132.8, 127.8, 118.3	103.1	58.3

Table 2.

Matrix	Fillers (Content)	ϵ_r	$\tan \delta$	Ref.
PP ^a	BT ^d (74 wt%)	8.8 @1 kHz	0.0044 @1 kHz	27
PP ^a	BT-Al ₂ O ₃ core-shell particles (13 vol%)	13 @1 kHz	0.10 @1 kHz	28
PP ^a	MWCNT ^e (2 wt%), BT (40 wt%)	15 @100 Hz	0.07 @100 Hz	29
PP ^a	rGO ^f (0.025 vol%)	12 @1 kHz	0.4 @1 kHz	13
PVDF ^b	ZnO ^g (17vol%), BT ^d (30vol%)	175 @100 Hz	0.43@100 Hz	21
PTFE ^c	T-ZnOw (30wt%)	5.1 @1 kHz	not reported	20
PP ^a	T-ZnOw (9.76 vol%), BT (15 vol%)	11 @1 kHz	0.04 @1 kHz	This work

^a polypropylene; ^b poly(vinylidene fluoride); ^c polytetrafluoroethylene; ^d BaTiO₃ nanoparticles; ^e multiwalled carbon nanotubes; ^f chemically reduced graphene oxide nanosheets; ^g radical ZnO nanorod; ^h tetra-needle-shaped ZnO whiskers.

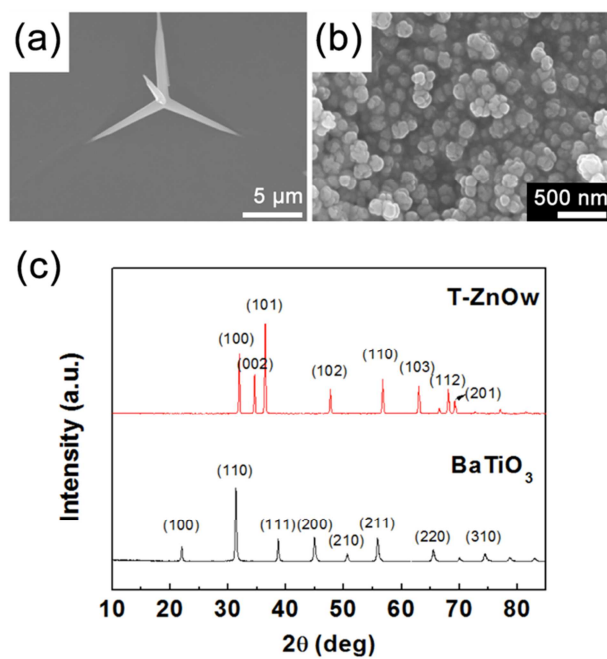
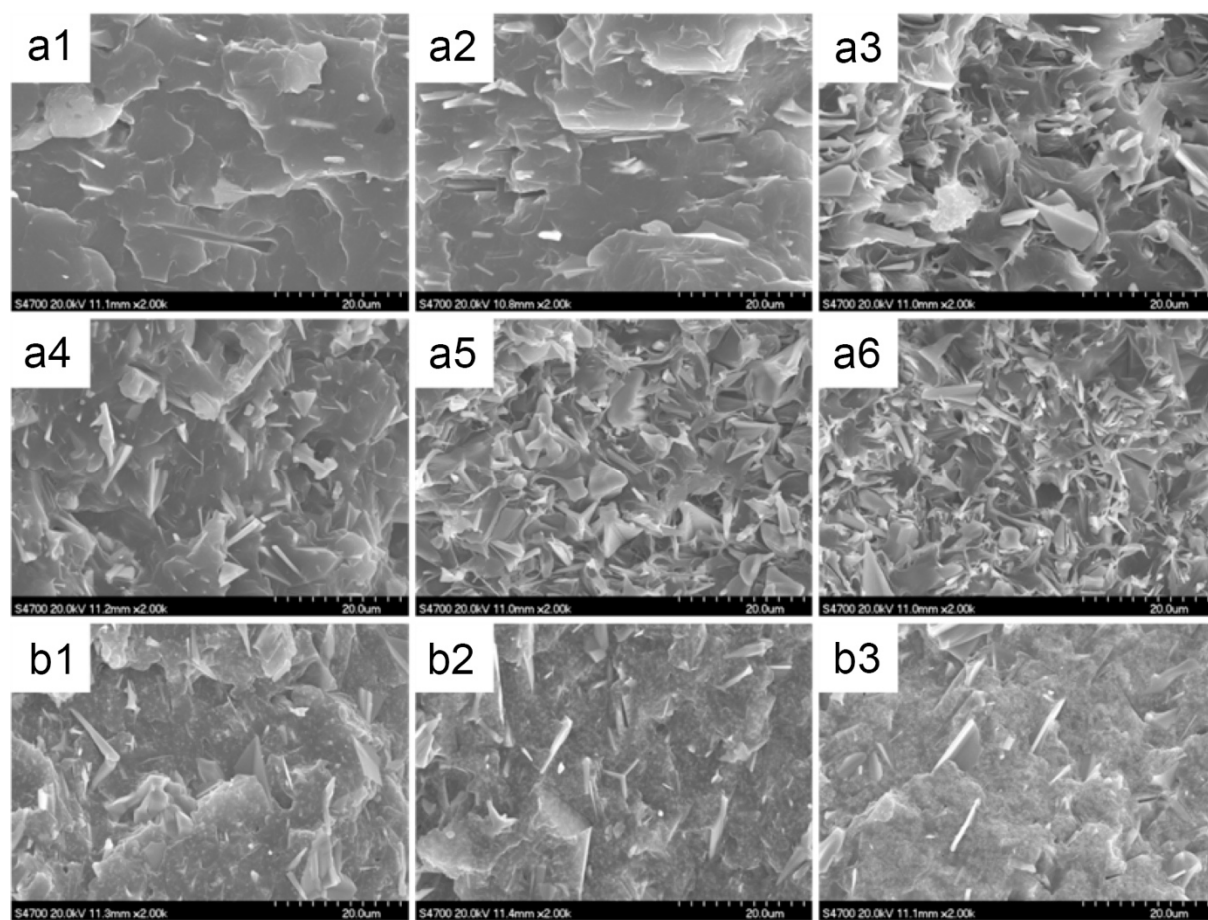


Fig. 1.

**Fig. 2.**

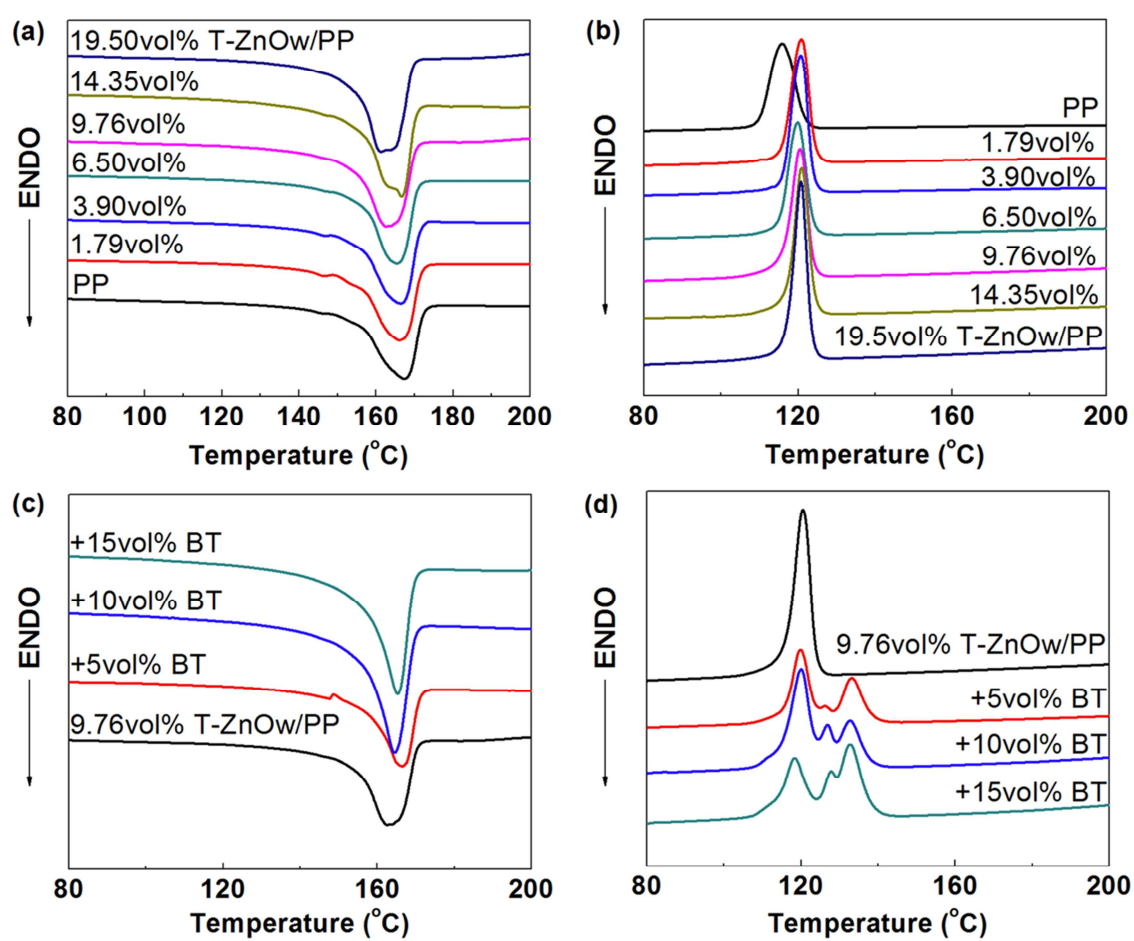
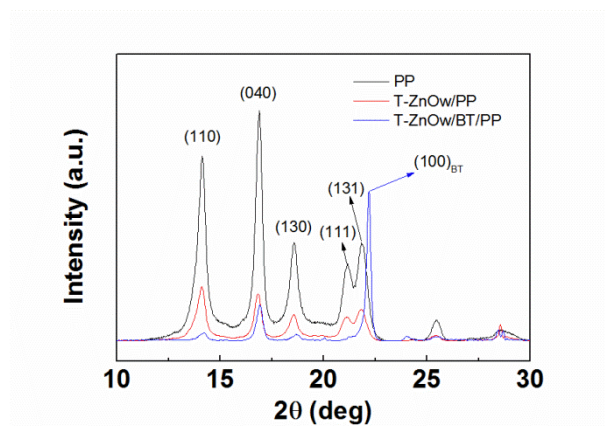


Fig. 3.

**Fig. 4.**

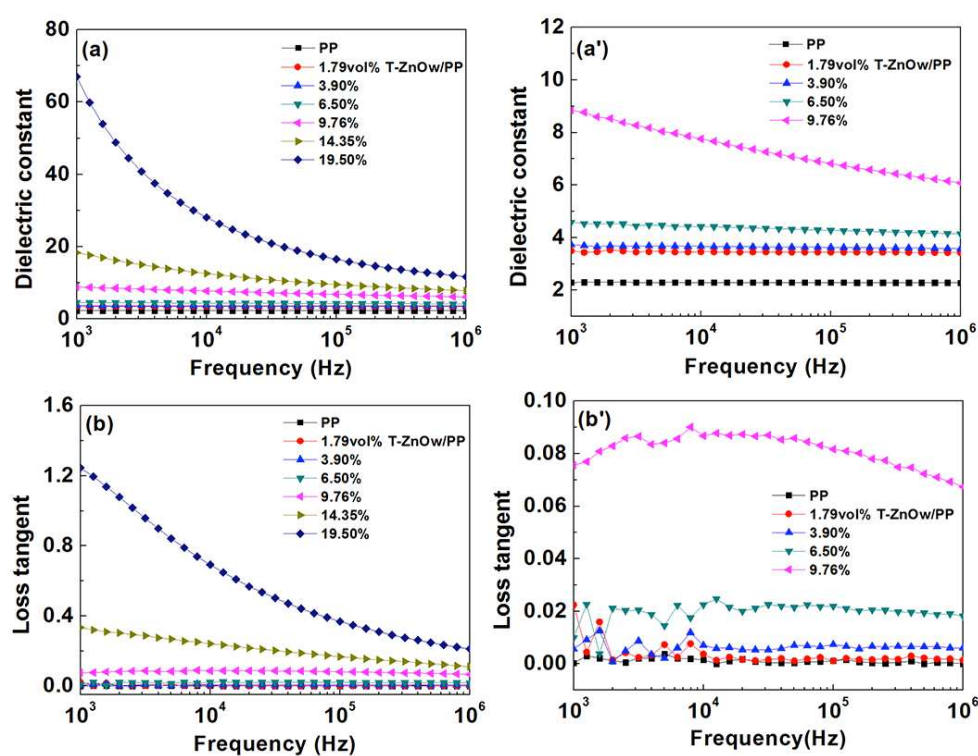


Fig. 5.

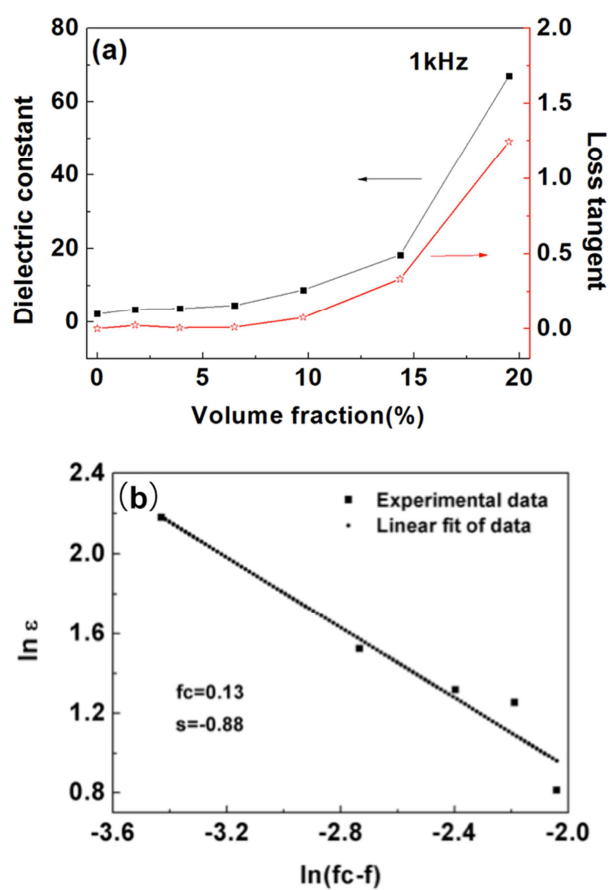


Fig. 6.

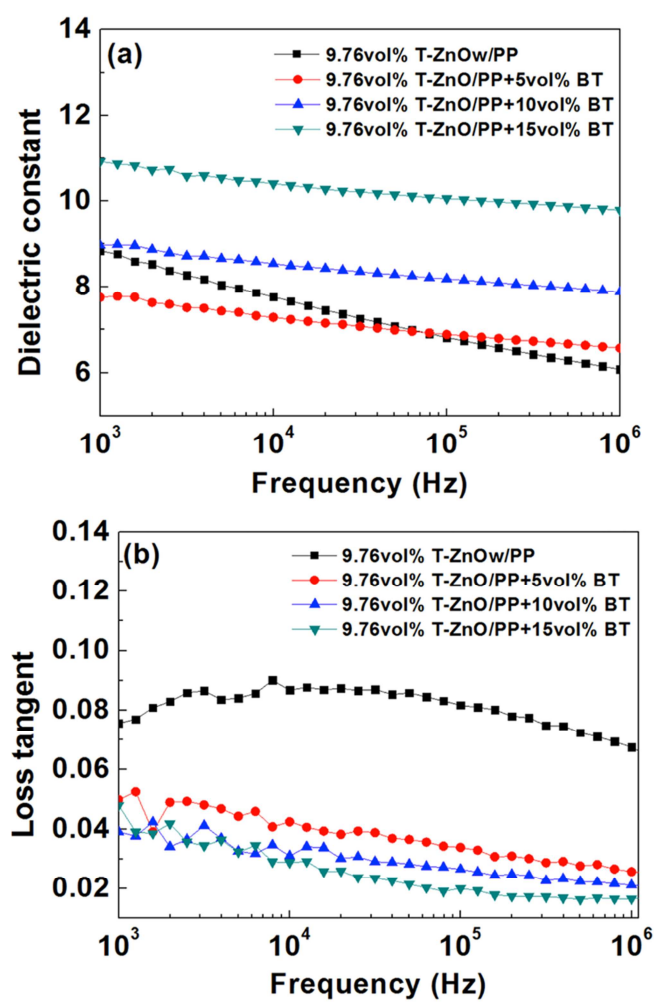


Fig. 7.